

The next generation of products and methodologies to establish biocrusts in degraded ecosystems

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Abstract

Biological soil crusts (biocrusts)—thin, coherent soil surface layers aggregated and inhabited by communities of tiny photoautotrophic and heterotrophic organisms—support ecosystem functioning in drylands and influence the Earth system. Biocrust loss due to disturbance, global change, and land use leads to altered hydrological function, susceptibility to erosion, diminished biogeochemical cycling, with consequences extending to regional and global environmental quality and ecosystem service provisioning. To restore degraded drylands, specific techniques could inform how to: 1. produce biocrust biota *ex-situ* for use as an inoculum in degraded environments, then 2. encourage the establishment of the inoculum in high-stress environments. To re-establish lost biocrusts, a promising pathway could be the integration of adding biocrust biota (inoculation), ensuring that it stays where applied (retention), and softening environmental stresses and limitations (amelioration) at approximately the same time. Several techniques, most of them developed in only the past few years, demonstrate this integration: biocrust sods, cloths, pellets, capsules, and slurries. Biocrust sods are fabric-reinforced patches of biocrust growing on soil, produced in farms, and

transported to and deployed intact in field sites. Biocrust cloths are fabrics and sheets with biocrust biota grown directly on their surface in greenhouse-based fog chambers that can be attached to soil in the field. Biocrust pellets are small, protective pieces of a binder material (often clay) infused with biocrust inoculum. Biocrust capsules are biodegradable gel capsules filled with biocrust inoculum, often with protective substances. Both pellets and capsules are broadcast over degraded soils. Biocrust slurries are semi-liquid suspensions of biocrust inoculum and other solids that are sprayed or spread in degraded sites. Emerging techniques like these have the potential to restore highly functional drylands and their ecosystem services. Potential applications include ecological restoration of dryland areas disturbed by fire, overgrazing, agriculture, mining and other factors, or environmental engineering in other contexts such as cultural heritage sites, croplands or human infrastructure. The next generation techniques for biocrust restoration described here are tools that could be applied to solve small- to medium-scaled challenges now, and with increased production efficiency and informed, strategic use of the products, large-scaled challenges in the future.

Impact Statement

Biocrusts are the living skin of drylands and pillars of the functioning of the ecosystems they inhabit, yet in many locations they are diminished by anthropogenic disturbances. Their loss can degrade drylands and, once degraded, their unassisted recovery may be orders of magnitude slower than their loss. Therefore, there is considerable interest and value in restoring biocrusts. In the past few decades, researchers have made progress in learning how to produce biocrust biota in dedicated biocrust cultivation farms (hereafter referred to simply as

'farms'), labs, greenhouses, etc., for use as inoculum. However, a remaining bottleneck is that field establishment of biocrust inoculum often fails or occurs more slowly than desired. Here, we review a new generation of emerging solutions and put forward a generalized recipe for supporting biocrust recovery: adding inoculum, retaining it in place, and passively mitigating environmental stresses and resource limitations. By optimizing these biocrust restoration methods, and developing others following the same principles, we can improve natural dryland and human environments.

Resumen

Las costras biológicas del suelo (biocostras) son capas superficiales delgadas y cohesivas del suelo, agregadas y habitadas por comunidades de pequeños organismos fotoautótrofos y heterotróficos. Apoyan el funcionamiento de los ecosistemas en las regiones áridas y semiáridas e influyen en el sistema terrestre. La pérdida de las biocostras, provocada por perturbaciones, cambios globales y uso del suelo, conduce a una alteración de la función hidrológica, mayor susceptibilidad a la erosión, disminución del ciclo biogeoquímico y efectos en cadena que se extienden a escalas regionales y mundiales, afectando el bienestar humano. Para restaurar las tierras secas degradadas, se necesitan técnicas que permitan: 1. producir *ex situ* la biota de las biocostras para su uso como bioinóculo en ambientes degradados; 2. facilitar el establecimiento del inóculo en condiciones de alto estrés. Para restablecer las biocostras perdidas, afirmamos que una vía prometedora es la integración de la adición de biota de biocostra (inoculación), asegurando que permanezca (retención) y suavizando las tensiones y limitaciones ambientales (mejoría) aproximadamente al mismo tiempo. Diversas técnicas, la

mayoría desarrolladas en los últimos años, demuestran esta integración: tapetes de biocostras, paños, pellets, cápsulas y lodos. Los tapetes de biocostra son parches reforzados con tela que contienen biocostras cultivadas en granjas, los cuales se transportan y despliegan intactos en los sitios de campo. Los paños de biocostras son telas y hojas con biota de biocostra que crecen directamente sobre su superficie en cámaras de niebla en invernaderos, y que posteriormente se pueden unir al suelo en el campo. Los *pellets* de biocostras son pequeñas unidades sólidas de materiales aglutinantes (a menudo arcilla), impregnadas con inóculo de biocostra. Las cápsulas de biocostra son cápsulas de gel biodegradable llenas de inóculo de biocosta, a menudo enriquecidas con sustancias protectoras. Tanto los *pellets* como las cápsulas se aplican sobre suelos degradados. Los lodos de biocostra son suspensiones semilíquidas de inóculo de biocostra y otros sólidos que se rocían o extienden directamente en los sitios degradados. Las técnicas emergentes como estas tienen el potencial de restaurar tierras secas altamente funcionales y sus servicios ecosistémicos. Las posibles aplicaciones abarcan zonas áridas perturbadas por incendios, sobrepastoreo, agricultura, minería y otros factores, pero también se extienden al entorno humano, incluidos los sitios del patrimonio cultural, las tierras de cultivo y el entorno construido. Las técnicas de restauración de biocostras de próxima generación aquí descritas son herramientas que pueden aplicarse para resolver los desafíos de pequeña y mediana escala actualmente, y, con una mayor eficiencia en la producción y un uso inteligente y estratégico de los productos, también resolverán desafíos de gran escala en el futuro.

Keywords

Biological soil crust, ecological restoration, ecological engineering, habitat amelioration, vegetation management

The problem, the promise, and the challenge

Biological soil crusts (biocrusts) are a thin, coherent soil surface layer aggregated and inhabited by communities of tiny photoautotrophic and heterotrophic organisms (Weber et al., 2022). These desiccation tolerant communities contribute key ecosystem functions in areas with low vascular plant cover, such as drylands, high latitude or altitude ecosystems, and disturbed mesic areas (Weber et al., 2022). Biocrusts cover an estimated 12% of Earth's terrestrial surface, across many major biomes (Rodriguez-Caballero et al., 2018). Biocrust composition can range widely in the dominance of key photosynthetic groups (i.e., cyanobacteria, algae, lichens, bryophytes) and can be strongly influenced by microsite conditions (Durham et al., 2018), climate (Ferrenberg et al., 2015; Phillips et al., 2022), disturbance regimes (Condon and Pyke, 2018), and subsequent succession (Weber et al., 2016). Of these factors, physical disturbances of the soil surface are a major constraint on biocrust development and distribution; biocrusts can be degraded or lost in hours to days during disturbance events but take years to decades (or more) to recover (Weber et al., 2016). Warming climates and shifting precipitation patterns can have similar negative effects (Ferrenberg et al., 2015; Phillips et al., 2022). Soil disturbance with associated effects on biocrust is expected to increase globally (Rodriguez-Caballero et al., 2018) due to high demands on croplands and rangelands, increasing needs for energy production, and urbanization (Olsson

et al., 2019). The loss of biocrusts results in the loss of their ecosystem functions, including soil stabilization (Phillips et al 2022) and erosion resistance (Rodriguez-Caballero et al., 2022), soil water cycling and storage (Duniway et al., 2018; Eldridge et al., 2020), building soil fertility and cycling nutrients (Rodriguez-Caballero et al., 2018), and influencing other biotic communities (Havrilla et al., 2019). Technology for restoring biocrusts is advancing rapidly, driven by the scale of environmental degradation in drylands and other biocrust habitats worldwide.

Biocrust restoration often begins with inoculation, the reintroduction of biocrust biota to sites where they have been lost or diminished. Inoculation reduces propagule scarcity as a barrier to ecosystem recovery (Bowker 2007). Propagules are materials that propagate the biota such as spores, tissue fragments, gemmae, or live cells, depending on the species. Biocrust inoculation approaches require a supply of biocrust propagules that can be added as inoculum. Researchers may use biocrust directly salvaged from the field, but techniques have been developed over the past ~20 years to first cultivate or culture this material into a larger quantity, hereafter termed “inoculum production” (Doherty et al., 2015; Antoninka et al., 2024). Inoculum production creates more material for use in restoration projects, allowing for larger scale treatments while decreasing or eliminating the need to obtain biocrust material from the field. Inoculum production has advanced significantly, especially in the last two decades (refer to reviews in: Palanački Malešević et al., 2024; Zhou et al., 2020). Production methods may target single isolates (Giraldo-Silva et al., 2019), populations of a target species (Doherty et al., 2018), or whole community cultures (e.g., Velasco-Ayuso et al., 2020; Antoninka et al., 2018). There are multiple production systems and modes, including liquid media cultures (e.g., Chamizo et al., 2020; Roncero-Ramos et al., 2019), fog chambers (e.g., Doherty et al.,

2020b), soil-based greenhouse growing beds (e.g., Velasco-Ayuso et al., 2020), and outdoor farming (Antoninka et al., 2024). In general, cyanobacteria and mosses have proven to be amenable to inoculum production (Zhao et al. 2016), but most lichens cannot be produced using existing techniques (Zhao et al. 2016), with a few exceptions (Bowker et al. 2017). Some of these methods have potential for up-scaled production, namely outdoor farming (Antoninka et al., 2024; Jech et al., 2023), repeated inoculation and harvest of solar fields (Heredia-Velasquez et al., 2023), and large liquid raceway ponds (Roncero-Ramos et al., 2019; Zhou et al., 2020). Different production targets and systems are appropriate in different settings and for different groups of biota, depending on the environment being treated, the goals and scale of the project and the type of biocrust to be regenerated.

Inoculum production is only one major bottleneck in the biocrust restoration process; another significant challenge is successful establishment of inoculum in the field. In our collective experience, we observe the best establishment results if we can achieve three objectives at or near the same time: **inoculation** (defined above), **retention** (retaining inoculum at an application site, reducing its loss), and **amelioration** (“softening” the environment; Figure 1).

Retaining inoculum on treated areas is challenging, with wind, water, or animals like ants moving materials off-site before establishment (Grover et al., 2022). Early applicators spread crumbled biocrust without amendments (Belnap 1993), relying only on fragment mass for retention and risking loss. Practitioners subsequently reduced losses by containing inoculum, fixing it to soil, or enhancing adhesion. For example simply watering spread inoculum aids soil adhesion (Antoninka et al., 2020; Ballesteros et al., 2017; Condon and Pyke 2016). Soil

surface imprinting is another retention practice (Doherty et al., 2020a). Biocrust inoculum successfully retained on site has an opportunity to establish, but must then overcome harsh environmental conditions that curtail or delay growth and establishment.

Amelioration of harsh environmental conditions can enhance biocrust establishment (reviewed in Palanački Malešević et al., 2024). Four such conditions include: 1. active sediment movement, 2. resource limitations, 3. abiotic stress factors, 4. biotic stress factors. Active sediment movement can result in inoculum loss from erosion, but also may affect biocrust inoculants through soil deposition and burial. Biocrusts have an organism-specific tolerance of burial, and partial burial may enhance establishment for some species (Jia et al., 2008; Martínez and Maun 1999; Williams and Eldridge 2011).

Water is the primary resource limitation in dryland biocrust habitats; biocrusts persist but cannot grow while desiccated. Therefore, successful multi-hectare applications have often included temporary irrigation (Zhou et al., 2020). In some habitats, nutrients may be equally or more limiting than water (Gretarsdottir et al., 2004). Other abiotic stressors include soil salinity or sodicity (Wu et al., 2022), UV exposure (Belnap et al., 2007), and temperature extremes (Bu et al., 2017). Biotic stress includes herbivory (Rinehart and Hawlena 2022), pathogens (Bethany et al., 2022), physical disturbance (Rengifo-Faiffer and Arana 2019), and competition (Ridenour et al., 2023). Vascular plants, especially exotic invasives, compete for resources and space while depriving biocrust of light through shading and litter (Detweiler-Robinson et al., 2013; Serpe et al., 2013). They also increase fire frequency and severity, causing direct mortality, though some biocrust survival has been observed when fire is the sole disturbance (Hilty 2004; Condon et al., 2023). Counteracting these stressors and limitations can increase restoration success.

The purpose of this review is to summarize recent progress in finding solutions to mitigate these barriers to biocrust restoration success. We emphasize approaches that simultaneously integrate **inoculation-retention-amelioration** (Figure 1) into a product or process. We also seek to catalyze expanded use of these products and processes in applications ranging from traditional ecological restoration to novel ecological engineering and sustainable infrastructure applications. Biocrust restoration with these methods may be deployed in wildlands to reverse degradation and boost recovery of ecosystem function or structure, including post-mining landscapes (Schultz et al., 2022), post-fire landscapes (Muñoz-Rojas et al., 2021), and abandoned roads (Chiquoine et al., 2016). However, biocrusts may also be deployed for direct societal benefits, such as conserving cultural artifacts like heritage structures (Cao et al., 2023). The industrial human-built environment provides multiple potential uses for these biocrust restoration tools and practices. For example, green roofs incorporating biocrusts boost biodiversity, retain water, and cool the human environment (Rocha et al., 2021). Another example is on roadside slopes and embankments where biocrusts are associated with greater soil stability or ecosystem development (Garcia-Palacios et al., 2011; Concostrina-Zubiri et al., 2019). Further, biocrusts can reduce wind erosion from bare soil beneath utility-scale photovoltaic installations (Wang et al., 2020). Restored or engineered biocrusts could act as natural fire barriers, slowing wildfires and protecting vulnerable areas by discouraging exotic annual grasses that carry fire (Muñoz-Rojas et al., 2021; Condon et al., 2023). With further testing, refinement and upscaling, inoculation-retention-amelioration methods for biocrust restoration may help us realize many of these applications.

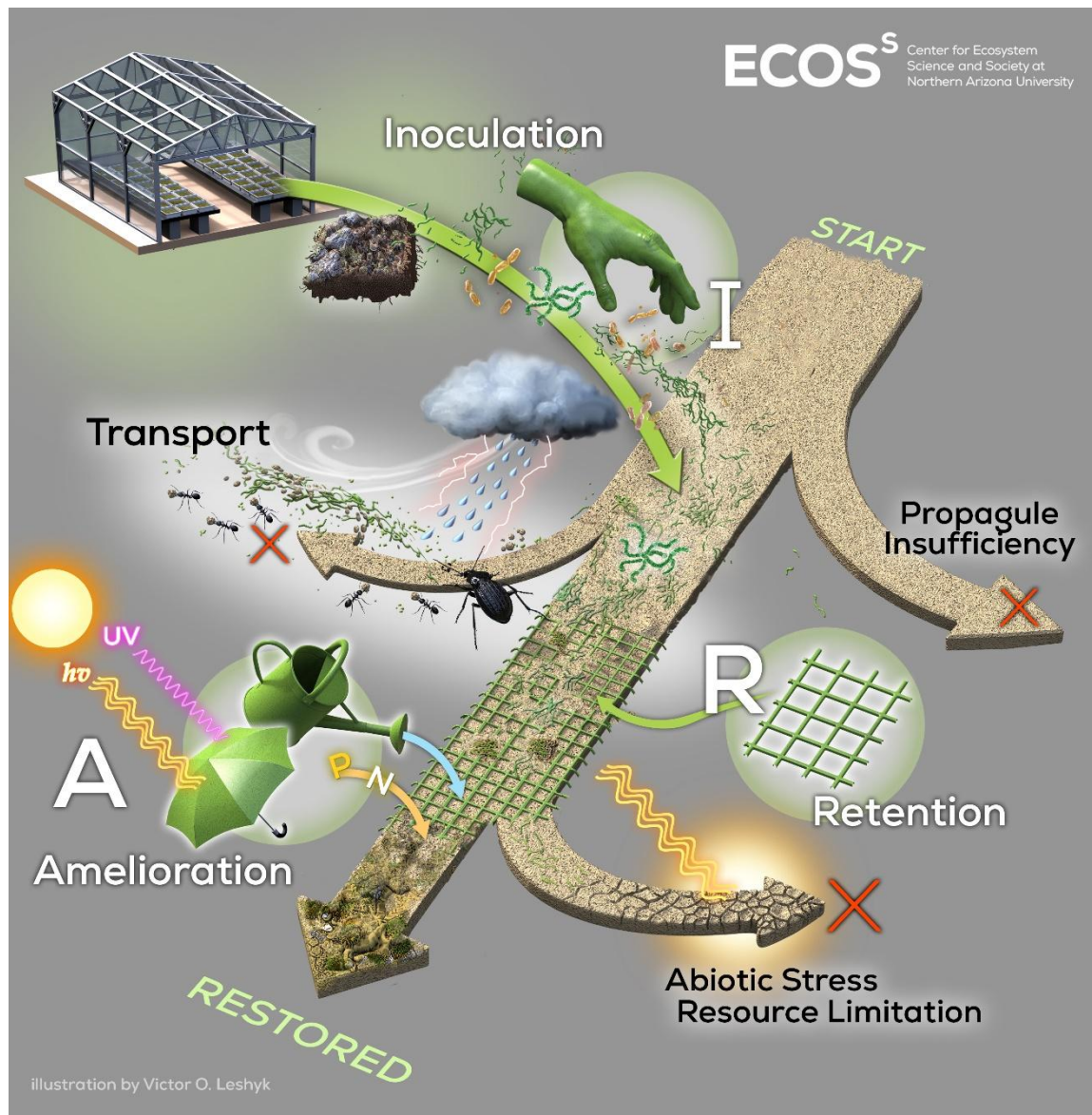


Figure 1. Illustration of the **inoculation-retention-amelioration** approach to biocrust restoration. Assisted biocrust recovery from disturbance can be thwarted by propagule insufficiency, transport of propagules off-site, or environment (abiotic stress and resource limitation). Inoculation is an approach to reduce propagule insufficiency. Retention is an approach to reduce losses of propagules to wind, water and animals. Amelioration is an approach to reduce abiotic stressors or enhance access to limiting resources such as water.

Bundling all of these strategies and integrating them into a single product or process is an approach to increase successful establishment (widen the arrow). This review presents emerging and developing techniques that adhere to this principle.

A classification of emerging inoculation-retention-amelioration solutions

In recent years a variety of potential new solutions have been advanced to assist the recovery of biocrusts in degraded ecosystems or anywhere their introduction might be beneficial, through implementing **inoculation-retention-amelioration**. These approaches rely on attachment of biocrust organisms to, containment of biocrust organisms within, or suspension of biocrust organisms in other materials. Once deployed, these approaches do not require additional inputs to maintain their retention or amelioration functions. These new approaches vary in the potential scale of use and intended outputs. Some lend themselves to broader applications, whereas others are targeted, small-scale interventions (Box 1). Because these approaches are relatively new or undergoing renewed research effort, they are not extensively covered in other recent synthesis products such as Faist et al. (2019).

Box 1. A key to emerging **inoculation-retention-amelioration** solutions to biocrust restoration.

1a. Biocrust delivered on or in solid media, not suspended in viscous semi-liquid

2a. *Biocrust growing on or reinforced by cloths, fabrics or sheets3*

3a. *Biocrust growing directly on soil, underlain by cloths, fabrics or sheets*

Biocrust sods (sod; Figure 2) are rollable and movable pieces of living biocrust. To date, they have been grown in outdoor farms, and though they can be variably sized, 1 x 3m has been a common dimension. Biocrust sods have a base-layer formed of a non-rigid cloth, fabric or sheet, overlaid by soil and finally the biocrust (a sod is ≥ 1 cm in thickness), with the optional addition of a shade/wind cloth surface layer. Sods can be unrolled over degraded soil surfaces to nearly

instantly create a patch of well-developed biocrust. Most intended applications of biocrust sods are small scale (e.g., treating erosional features, trail or road decommissioning, etc.), although some strategic uses could be envisioned that have large scale effects (e.g., creation of a firebreak). Sods can be created using different materials for different desired outcomes, the most essential being the base-layer.

Variations:

Plant-permeable biocrust sods are specifically intended to establish biocrusts without excluding, and possibly encouraging, vascular plants (particularly native plants). These could be used when both enhanced native plant and biocrust abundance is desired to meet restoration or other management goals (e.g., control of soil erosion). Typically, in this type of sod, the base-layer is a woven organic cloth (burlap has been commonly tested, but other possibilities exist). The weave must be tight enough to contain biocrusted soil, but also be penetrable by roots growing from above or shoots growing from below.

Weed-blocking biocrust sods are specifically intended to establish biocrusts while discouraging vascular plants (particularly exotics). These could be used when high prevalence of exotic plants discourages biocrust development, or when low vascular plant abundance is desired to meet management goals (e.g., some photovoltaic arrays, fire-breaks). Typically in this type of sod, the base-layer is one of various biodegradable weed barriers, such as those made from maize-derived polylactic acid (PLA), though other options exist. Another layer of a reinforcing organic cloth (e.g., burlap) may also be present atop the base-layer.

3b. Biocrust growing directly on cloths, fabrics or sheets with no or minimal soil
Biocrust cloths (Figure 3) are pieces of organic cloth or other materials covered in or infused with biocrust organisms. This product has primarily been tested specifically with biocrust mosses, with the growth occurring in fog-chambers. The cloth material provides a medium for biocrust attachment that can be applied to soil surfaces with the biocrust-covered surface oriented downward to shelter the biocrust. Biocrust cloths have the advantage of being lightweight, and can easily be cut to various sizes either before or after growth of the biocrust organisms. Most intended uses of biocrust cloths are small scale and targeted, though if strategically placed could have medium scale outcomes.

Variations:

Plant-permeable biocrust cloths are intended to establish biocrusts while having neutral or positive effects on vascular plants. To date, this type of biocrust cloth has been created using burlap cloth, which has a loose

weave that allows vascular plant seedlings to grow through from above or below.

Weed-blocking biocrust cloths establish biocrusts while discouraging vascular plants. To date, maize-derived PLA sheeting has proven to be the most effective material.

2b. Biocrust contained or embedded within another solid material in small, broadcastable units

Biocrust pellets and capsules are meant to be dispersed, either by hand or using machines (such as farm equipment and drones) at small to large scales. To date, all tested examples have been $\leq 1\text{cm}^3$. Pellets and capsules physically shield biocrusts from abiotic and biotic stressors in their environment while the inoculum is gradually released. Both pellets and capsules can also contain other biological materials in addition to biocrust biota, such as soil microbiota or seed. Both can also contain various adjuncts that might benefit biocrust establishment by repelling herbivores, boosting stress tolerance or providing/retaining resources.....4

4a. Biocrusts are embedded within pieces of mineral or organic media

Biocrust pellets (Figure 4) incorporate biocrust organisms into structures aggregated together by minerals or binders. Clay is the most common mineral binding agent because it can be mixed into a dough-like medium; however, pellets based on diatomaceous earth and other materials have also been tested. Through time, pellets dissolve through wetting cycles, releasing biocrust propagules. Clay pellets in particular may retain moisture, enabling survival and encouraging growth of biocrusts even before the pellet has disintegrated.

4b. Biocrusts are contained within gel capsules

Biocrust capsules (Figure 5) are made by filling gel capsules, like those used for medicines, with biocrust biota. Gel capsules are made of various materials, all designed to dissolve with water or acidity. These also dissolve, crack or disaggregate when exposed to the elements outdoors. When wet, they may partially attach to the soil surface, anchoring them in place. Most capsules are at least partly transparent, potentially allowing biocrust photosynthesis to occur inside of the capsules.

1b. Biocrusts are suspended in a viscous semi-liquid

Biocrust slurries (Figure 6) are made by mixing and suspending biocrust biota in water prior to application. Unlike wet applications of growth media containing biocrust biota, slurries contain some suspended tissue or soil solids and/or adjuncts such as tackifiers that promote adherence to the soil surface or protection of the biota from environmental stressors.

History and state of the art

Biocrust sods

History

Biocrust sods originated from an outdoor cultivation method where practitioners spread thin layers of inoculum over soil between fabric layers and allowed it to grow (Jech et al., 2023). These "quesadillas" (an analogy to a food in which a thin layer of fillings is layered between thin flatbread) are multi-layered assemblies atop a base of synthetic weed cloth, where the sod itself consists of a bottom layer of organic biodegradable cloth (sometimes multiple types), a thin soil layer inoculated with crumbled or sieved biota and a top layer of shade cloth, all drip-irrigated. This method enhances growth by increasing moisture retention, reducing UV exposure, preventing wind loss of inoculum, and moderating temperature. These favorable conditions allow biocrust to remain active for weeks, as opposed to biocrust in situ which spends significant amounts of time dormant, resulting in a well-developed crust in less than a year. Early harvest consisted of scraping the cultivated material from the sod, but a key advance was utilizing the entire sod in the restoration process as detailed below. Sods are grown in long rows (up to 30 m), approximately 1 m width. At harvest the rows are cut to 3 m length and rolled into tightly packed cylinders for transport to the target site, where they can be unrolled directly on the application surface while maintaining greater intactness compared to scraping and crumbling (Figure 2 a-c). As opposed to scattering inoculum, installing biocrust sod creates an instant island of robust biocrust (**inoculation**) attached to fabric-reinforced soil (**retention**), reducing local soil erosion and weed pressure (**amelioration**) (Figure 2 d, e). We note that the biocrust sod bears a resemblance to the hypothetical "pan-life-carpet" proposed by Palanački

Malešević et al. (2024), demonstrating that other researchers are converging on similar restoration solutions.

From 2017 to 2019, an outdoor biocrust farm in southern Utah produced about 6,000 m² of biocrust from multiple local and non-local sources, harvesting a portion as sods (Faist et al., 2019). Sods were first field tested in southern Utah, installed in the field with or without shade cloth. One year post-installation, moss and lichen coverage was <1% in untreated locations but increased to ~3% with unshaded sods (both local and non-local sources), and ~17% with locally-sourced sods with shade cloth. After 5 years, biocrust cover ranged from complete loss in heavily grazed areas to as much as ~80% moss cover in shaded, locally-sourced sods (Tucker et al. 2025). Sod application also reduced exotic cheatgrass (*Bromus tectorum* L.) cover to <1.2%, one-third the coverage in untreated controls (Jech et al., 2023).

In 2019, researchers compared biocrust establishment of biocrust cultivated in three ways at a Sonoran Desert farm; the three ways were in quesadillas (where the shade cloth sits directly on the inoculum), in hoopouses (holding shade cloth one meter above the cultivation substrate), and in as greenhouse (Antoninka et al., 2024) Each inoculum was then applied in the field as both small sods and as loose inoculum after scraping the cultivated biocrust off the fabric (Rowe et al., 2025). Outdoor production methods produced inocula that established better than greenhouse-produced inoculum and allowed an improved opportunity to germinate and eliminate weeds prior to deployment (Antoninka et al., 2024). The quesadilla method best retained source community composition (Antoninka et al., 2024). Sods outperformed scattered inoculum over soil but performed similarly to scattered inoculum over burlap, demonstrating burlap's retentive and ameliorative value (Rowe et al., 2025). Overlaid

biocrust sods suppressed native seeding success, though seeding under burlap alone was beneficial (Rowe et al., 2025).

Recent developments & innovations

More recent findings suggest biocrust sods can be used to combat erosion. Olney (2025) installed plant-permeable biocrust sods perpendicularly across small catchments, upslope from erosional features in the Sonoran Desert. Beyond attaining high biocrust coverage associated with well-developed biocrusts (>40% initial coverage of dark cyanobacteria, moss, and lichens), this coverage more than doubled during the one-year study, indicating field growth even with below average precipitation. Treated plots had several times the coverage of untreated controls after two growing seasons. Furthermore, sod placement increased soil surface aggregate stability between sods and reduced sediment flux during some monitoring periods (Olney 2025).

The authors have explored recent refinements of biocrust sods to include innovations in materials and farming methods (e.g., Olney 2025). Combining multiple fabrics, such as PLA weed barrier overtopped by burlap cloth, can confer weed-resistance and enhanced structural soundness. Limiting soil to 0.5-1cm thickness versus earlier iterations with ≥ 2 cm results in lighter sods which are more compact and easier for two people to carry. Moistening sods on transport days makes them more pliable for rolling with less damage if field-installed within about one day; however, they should not remain rolled and hydrated for extended periods or exposed to hot temperatures while wet.

Outdoor biocrust farming succeeds best in cool seasons (ideally $< 19^{\circ}\text{C}$; Condon and Pyke 2016, Fisher and Subbotina 2014), creating logistical challenges with alignment of the

completion of farming and ideal application times. Once daytime highs consistently reach 27°C, it is best practice to stop irrigation to minimize stress, curtailing the growing season and creating fallow periods. Ongoing trials by the authors indicate that ceasing summer irrigation and resuming when weather cools may stimulate additional successional development, producing active, vigorous biocrust synchronized with ideal fall installation windows rather than months of desiccated storage. Another solution could exploit climate differences among multiple farms. Growing low elevation-sourced biocrust at cooler, high elevation farms during warm seasons may prove quite productive and produce well-timed harvests for low-elevation deployment, with preliminary observations by the authors suggesting sufficient field establishment. Multiple biocrust sod farms on elevation gradients could theoretically supply two or more harvest periods annually.

Before introduction into field sites, we must ensure sods are weed-free. In the production phase, sods are kept moist to maximize biocrust growing time which can trigger the germination of seeds sourced either from the original field-collected biocrust starter or from the farm site. Triggering weed germination could help exhaust a pre-existing seed bank within the biocrust, but only if emerging weeds are prevented from setting seed. Manual control of weeds can be expensive, labor intensive, and disturb the biocrust surface. Growing sods may be treated with herbicides to greatly reduce weed growth, without preventing biocrust development (Rowe et al. in press). Some herbicides begin losing their toxicity to native plants at sod harvest, meaning that treated sods need not introduce active herbicides into the field unless desired (Rowe et al. in press).

Remaining challenges & opportunities

There are remaining practical challenges in sod production. One is that during the production of plant-permeable biocrust sods (with a burlap base), the burlap may decay to a degree that could hinder rolling and transporting the sod for deployment elsewhere. Weed-blocking sods that feature an extra base layer (biodegradable weed barrier) may not degrade as quickly, but are specifically designed to not be permeable to plants once deployed. Alternative bottom layers could be identified that outlast burlap alone, yet are permeable to vascular plant emergence shortly after deployment.

There are opportunities to increase the spatial scale of outcomes of biocrust sod use through their arrangement and grouping in space. A single biocrust sod creates only a patch of restored biocrust and erosion protection, and in some cases, exotic plant reduction. However, the modularity of the units means that practitioners can arrange sods to serve their goals, such as linking them together end-to-end to create a fire break or to stabilize a hillslope on contour, or grouping them to create restoration islands along with integrated seeding or planting. Additional tests can help determine how best to use sods to boost native plant establishment.

In these early trials and in ongoing work by the authors, biocrust sods have always been initiated with natural biocrust material, salvaged prior to a planned disturbance. When added at the initiation of sod farming, all major groups of the biocrust biota have been observed to survive and often grow. Cyanobacterial growth rates are generally the fastest, followed by mosses, and both can attain a high coverage. Lichens may grow during sod farming, but less quickly and less consistently. It remains to be determined if biocrust sods can be created with *ex-situ* cultured biocrust biota, rather than field-sourced material.

The potential utility of biocrust sods is not limited to degraded dry wildlands. They could be used in a variety of settings in the human environment such as rejuvenating cropland soils, containing pollutants, or creating green roofs in dry to mesic regions (Rocha et al., 2021).

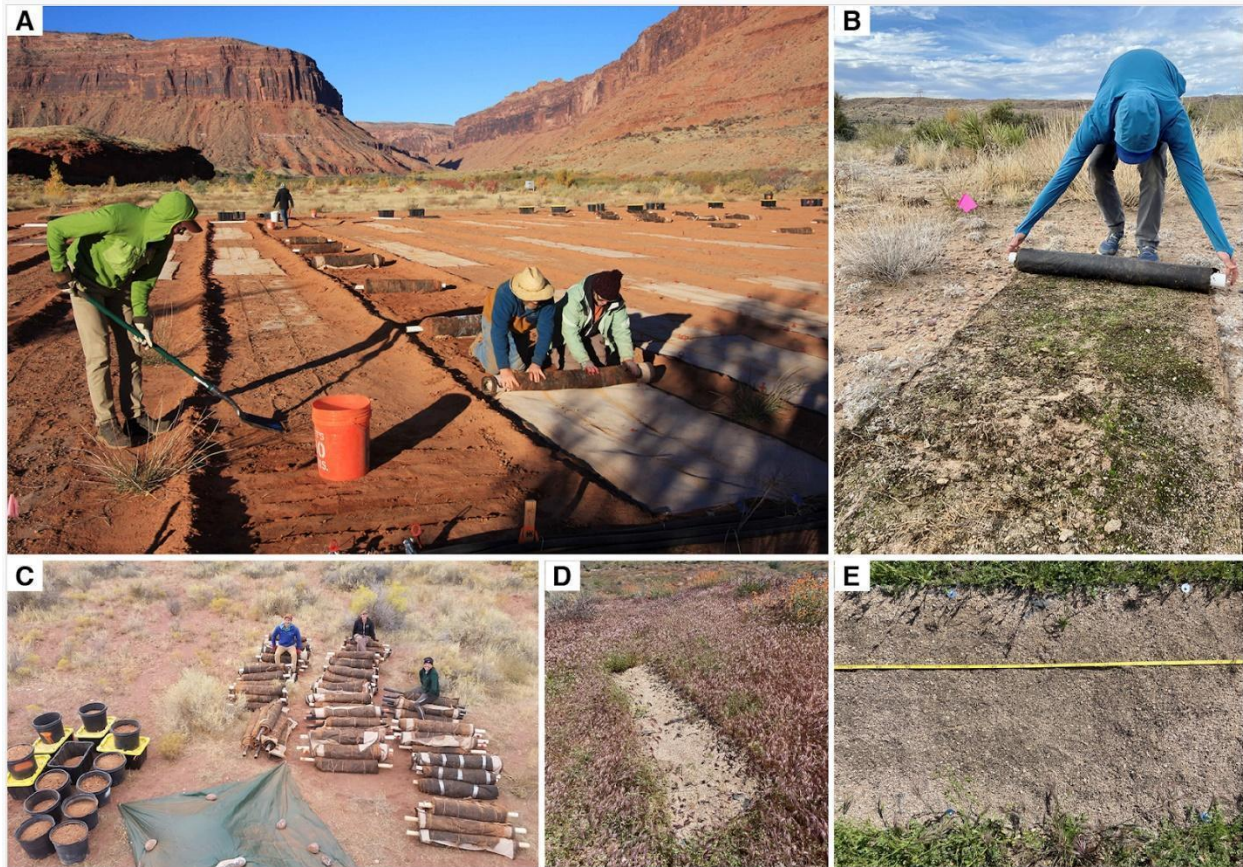


Figure 2. Production, harvest and use of biocrust sods. A. Harvesting and rolling sods at the southern Utah sod farm; B. Unrolling a sod in the Sonoran Desert. Sodds were wetted before harvest to increase cohesion when rolling, transporting, and unrolling; C. Harvested rolls of sod and buckets of crumbled biocrust at the southern Utah farm; D. A biocrust sod with a corn-derived, biodegradable weed barrier base layer plus a layer of burlap, shown here surrounded by the exotic grass *Bromus rubens* L. in the Sonoran Desert; E. A biocrust sod in the Sonoran

Desert three months after initial installation. Photo credits: Colin Tucker (A & C); Keven Griffen (B, D, & E).

Biocrust cloths

History

Fabrics and geotextiles have often been used as restoration and rehabilitation tools to reduce runoff and soil erosion (Krenitsky et al., 1998) and establish vegetation (Tamura et al., 2017). Fabrics can ameliorate the environment through shading, protecting plants from freezing conditions, and increasing available moisture similarly to mulch (Bainbridge 2007; Kertész et al., 2011). Based on the success of fabrics for vascular plant restoration, researchers began testing similar methods for combining biocrust inoculation with placement of fabrics including jute net (Condon and Pyke 2016; Slate et al., 2020; Bowker et al., 2020), burlap (Doherty et al., 2020b), and weed fabrics (Bowker et al., 2025). Condon and Pyke (2016) first tested the ameliorative efficacy of fabrics (jute net) on biocrust inoculum (salvaged moss) and found that it increased cover and establishment. Slate et al. (2020) confirmed the efficacy of the technique in conjunction with moss inoculum addition. Plant-based fabrics, including jute net, burlap, and similar materials, have also been shown to facilitate the establishment and increase of cyanobacterial and mixed inoculum in diverse settings on multiple continents (Bowker et al., 2020; Zhao et al., 2021; Roman et al., 2021).

An early study pioneered a distinct usage of fabrics in biocrust restoration, infusing small pieces of hemp cloth with cyanobacteria as a delivery mechanism for biocrust inoculum to the field, resulting in a short-term increase in the abundance of cyanobacteria but ultimately no

long-term effect (Kubečková et al., 2003). Bringing together the infusion of inoculum with cloth (Kubečková et al., 2003) and the demonstrated ameliorative effects of fabrics in the field (Condon and Pyke 2016) resulted in a greater degree of success in moss biocrust establishment. Doherty et al. (2020b) developed a cultivation system for mosses using a chamber equipped with fog emitters producing sub-micron water droplets, circulating across multiple layers of burlap fabric substrate (Figure 3). The authors cultivated the moss *Syntrichia ruralis* (Hedw.) F. Weber & D. Moh on burlap, achieving a 17% biomass increase over 2.5 months. The fog system enables space-efficient cultivation, with the potential for stacking multiple layers of burlap vertically to maximize production. This produced moss-colonized burlap sheets, which can be attached to soil surfaces (Doherty et al., 2020b). Using these “moss cloths”, a high degree of establishment (~60% cover) was achieved over two growing seasons, relative to free-living wild and cultivated materials, especially when the biocrust cloth was applied with the moss-covered side facing the soil. This method may be successful because the moss propagules (**inoculation**), are both attached to the cloth material and held underneath it against the soil (**retention**), and the cloth material provides a cooler, moister growing environment fostering establishment (**amelioration**).

Recent developments & innovations

Recently, researchers have explored variations of the moss cloths described above, altering the materials used, and exploring new uses for the established technique (Doherty et al., 2020b). One study investigated the efficacy of biocrust moss cloths as a weed barrier, with the goals of simultaneously reducing exotic plants and establishing biocrust propagules (Bowker et al., 2025). *S. ruralis* was grown over burlap and four new weed barrier fabrics ranging in

durability and composition, then patches of the various cloths were field tested. In addition to burlap, a maize-derived, biodegradable PLA weed fabric also promoted moss biocrust field establishment in addition to reducing weed abundance over one growing season, but not long-term (Bowker et al., 2025). Bowker et al. (2025) suspect this is the case because the weed barrier did not persist long enough in the environment. Thus, a longer-lasting biodegradable weed barrier might be the key to simultaneous biocrust establishment and longer-term weed reduction.

Given the effectiveness of burlap moss biocrust cloths in establishment of biocrust mosses, Anenberg (2025) investigated whether the cloths might also facilitate the establishment of seeded herbaceous vascular plants through habitat amelioration and crossed experiments with rodent exclusion. Burlap (without moss), burlap moss biocrust cloth, and moss detached from burlap all similarly boosted the success of seeding compared to seeding alone, in the absence of rodents. However, in the presence of rodents, the cloths had no effect on seeding success. The presence of the rodents reduced the residence time of the burlap on the soil surface. Thus, moss cloths could plausibly be deployed both to boost biocrust cover and establishment rates of seeded vascular plants under ideal conditions, but rodent pressure remains a barrier to overcome.

Remaining challenges & opportunities

At this time, the biocrust cloth methods described above have mostly been tested using a single moss species, and we do not yet know if other types of biocrusts could be established in a similar way. Cooke et al. (2024) tested the broader applicability of fog-chamber inoculum production for additional species. In addition to *S. ruralis*, the moss *Homalothecium aureum*

(Spruce) H. Rob. was demonstrated to grow under fog-cultivation, but two lichens (*Cladonia* spp., *Peltigera rufescens* (Weiss) Humb.) did not (Cooke et al. 2024). Kebede et al. (2023) also found that a fog-cultivation system produced positive growth of three additional biocrust mosses: *Ceratodon purpureus* (Hedw.) Brid., *Bryum argenteum* Hedw., *Funaria hygrometrica* Hedw. Though neither study produced biocrust cloths, these results suggest that cloths covered with any of these moss species could likely be produced, opening the door to a multi-species product and restoration of biocrust communities. So far, researchers have been unable to extend the technique to lichen biocrusts and have not explored it for cyanobacteria beyond very small cloth fragments.

The authors have also observed fungal contamination of cultures, which may affect growth and viability of the biocrust cloths. While the origin of these fungi is not clear, likely candidates could include residues from wild harvested moss or perhaps spores from the greenhouse facilities in which fog chambers are situated. Cultures may require a combination of fungal decontamination, preventative fungal control treatments, and adaptations to the chamber design, which are still being examined.

Perhaps the greatest challenge and an interesting opportunity is learning how to integrate the use of biocrust cloths with vascular plant-based restoration practices. Already, there is some evidence that biocrust cloths can modify the success or failure of both exotic and native plants (Bowker et al., 2024; Anenberg 2025), but we know very little beyond small test patches in a few sites. Possibly, biocrust cloths could provide the erosion control, or weed-free window that native perennial plants require to establish on eroding slopes, burned environments, herbicide-treated areas and other contexts. Techniques exist to facilitate plant

establishment in such environments (e.g., herbicide use, geotextiles, etc.), but biocrust cloths may provide the additional benefit of biocrust establishment and thus a potential long-term, self-maintaining weed- and erosion-deterrent.

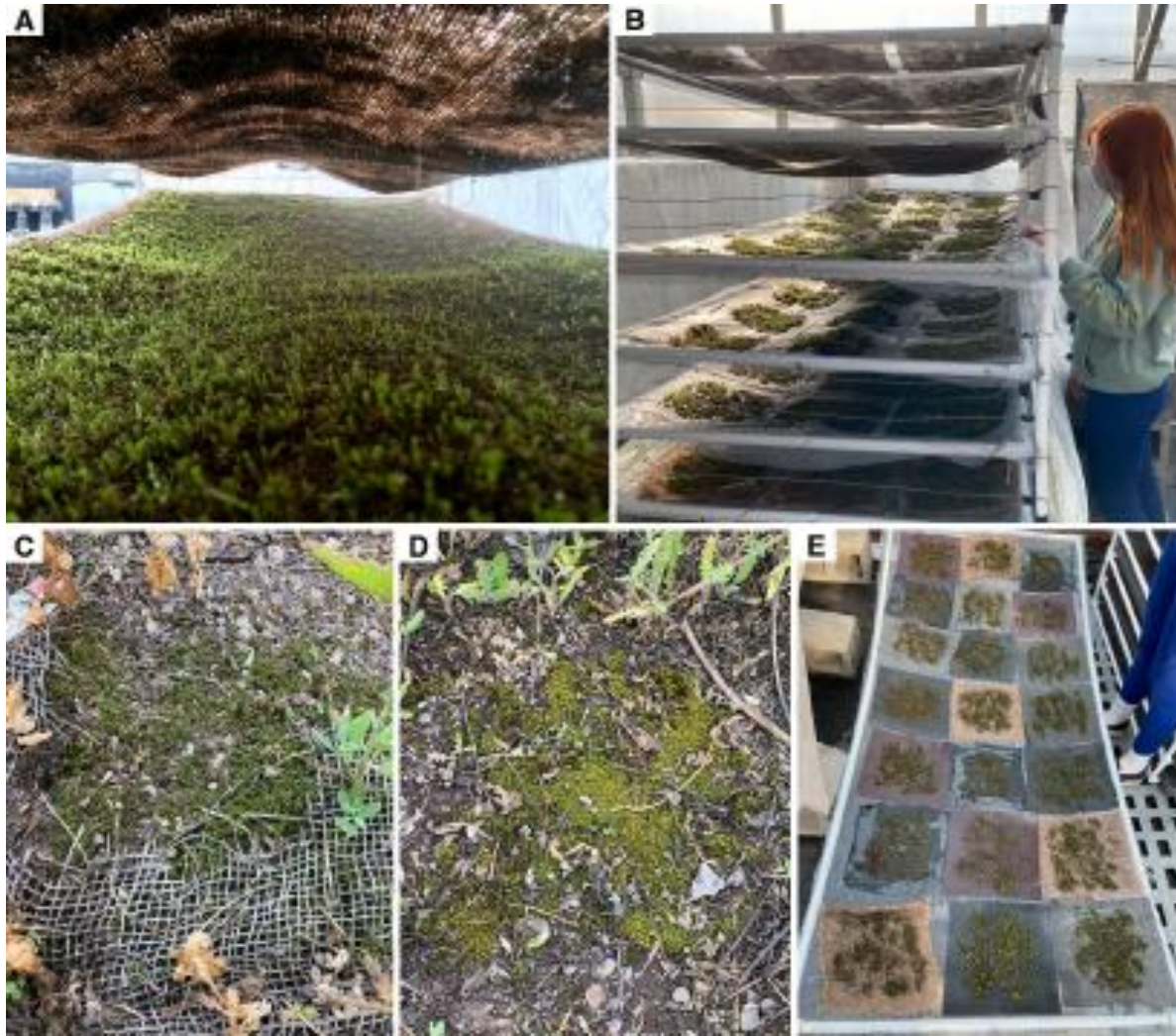


Figure 3. Production and use of biocrust cloths. A. *S. ruralis* growing on a burlap base in a greenhouse fog-chamber, where the moss accumulates biomass and attaches to the cloth; B. A multi-level system of racks irrigated with fog (fog-retaining row cloth cover not shown) is used to produce moss cloths. C. Field-established *S. ruralis* burlap cloth, shown after surviving one summer season with mosses emerging through remaining cloth fibers (herbaceous vascular

plants are incidental colonists); D. A patch of high moss cover established using biocrust cloth in which the cloth is no longer detectable; E. Squares of biocrust cloths made of the moss *Syntrichia ruralis* grown on a variety of bases, including burlap and various weed barriers. Photo credits: Jasmine Anenberg.

Biocrust pellets

History

Seed pellets in restoration trace back to the 1940s, when seedlings emerging from rabbit droppings inspired seed-infused clay pellets for aerial seeding (Jordan et al., 1967; Roundy 1995). Seed pellets are variously called "balls", "bombs", "pods", "pillows", "pasta", and "nendo dango" (clay dumpling), but all are based on seeds mixed with binders formed into shapes by hand, using rotating drums, or with pasta machines (Gornish et al., 2019; Kannan et al., 2021; Madsen et al., 2016a). Seed pellets aggregate seeds within another medium, most often bentonite clay, though pectin, psyllium, xanthan gum, cellulose, and polyvinyl acetate glue have been used (Liu et al., 2020). Pelletizing can protect seeds from herbivory and wind/water transport while aiding dispersal in aerial seeding (Chadwick et al., 1969; Gornish et al., 2019).

Seed pellet methodologies could be adapted for biocrust restoration. Seed pellets are often infused with adjunct ingredients to reduce stress and enhance resource access (Madsen et al., 2016b; Madsen et al., 2014). Absorbent ingredients such as clay, pectin, and synthetic hydrogels help reduce water stress. Growth-promoting adjuncts include plant hormones like abscisic acid, minerals such as azomite and limestone, and organic matter including biochar,

compost, and fertilizer (Anderson et al., 2021; Richardson et al., 2019). Diatomaceous earth has protected seed and biocrust propagules from ant predation (Grover et al., 2022). Other additives, such as chili powder or oils, could protect against mammals (Taylor et al., 2020).

Early biocrust pelletization began with cyanobacteria mixed with coagulating agents like starch or alginate; though lab and greenhouse trials were promising, later field trials were less successful (Buttars et al., 1998; Buttars et al., 1994; Howard and Warren 1998). Modern seed pellet methods were adapted by Roman et al. (2020), who created extruded clay and cyanobacteria pellets resulting in enhanced cyanobacterial colonization in a greenhouse. Liu et al. (2021) found peat pellets successful for growing biocrust moss but did not intend them as restoration tools. Grover et al. (2022) created diatomaceous earth pellets infused with moss tissue to protect propagules from ant herbivory in post-fire contexts, finding pelletization increased moss establishment. A later test of this method indicated a short-term benefit for mixed biocrust establishment in the Sonoran Desert (Antoninka et al., 2024). These findings sparked inquiry into adapting pellet technology for biocrust restoration, as pellets may deliver biocrust inoculum (**inoculation**), increase inoculum residence time or soil adherence (**retention**), and include additives that enhance resource access or blunt stressors (**amelioration**).

Recent developments & innovations

Biocrust pellets containing the moss *Syntrichia ruralis* were tested in the sagebrush steppe of Montana to identify if pellet size, shape, and content alters moss establishment compared to inoculation with scattered moss fragments (Anenberg 2025). All pellets were clay based and used the same moss inoculum, previously cultivated in a fog chamber. The pellets

were formed into four combinations of size and shape, and two different formulations of pellet contents were used. One formulation contained more cellulose and no pectin as adjuncts, whereas the other included pectin and less cellulose. After two years in the field, some but not all pellet types established more moss than scattered moss fragments, with the single best performing pellet being small, round, and containing pectin. Pectin inclusion was the most important positive factor for increasing moss establishment. This study demonstrates that biocrust pellets can in fact lead to greater biocrust establishment, but serves to illustrate that not all pellets are equally effective (Anenberg 2025).

Co-delivery of biocrust biota (*S. ruralis* tissue) and vascular plant seeds (*Achillea millefolium* L.) in clay-based pellets containing hydrogel polymer was tested in a greenhouse over irrigated sand beds, compared to unpelletized controls. Inclusion of seeds or biocrusts in pellets *decreased* their establishment. This may have occurred because pellets are designed to enhance retention and ameliorate stressors, but these benefits may not materialize in controlled, low-stress environments. Any hindrance to biocrust or seed viability from pellet manufacture or sub-optimal composition could thus express as negative pelletization effects. A significant finding was that when both were incorporated into pellets, *S. ruralis* positively affected *A. millefolium* establishment, while *A. millefolium* negatively affected *S. ruralis* establishment (Antoninka et al., 2022). Though we cannot determine if different biocrust-seed combinations would yield similar patterns, the findings suggest species interactions play important potential roles in biocrust-seed pellet use.

Subsequent field studies of biocrust establishment from pellets failed to achieve the Montana trial's success (Anenberg 2025), indicating much remains to be learned. Clay-based

pellets (16 types) containing three post-fire biocrust moss species (*Ceratodon purpureus*, *Funaria hygrometrica*, and *Bryum argenteum*) were deployed into an experimental burn area in northern Arizona, and the authors observed a lower biocrust cover from pellets compared to scattered inoculum (Bowker et al., 2023). In another Sonoran Desert test, cyanobacteria and native plant seeds in clay-based pellets failed to increase biocrust cover after one season but slightly boosted seeded plant emergence (Antoninka et al., 2025). While these trial failures are difficult to explain, possible factors include insufficient biocrust inoculum quantity, incompatibility of certain taxa with pelletization, unfavorable weather, or unfavorable propagule release timing.

Oliveira et al. (2026) have very recently developed an alternative pellet-like method: transparent alginate beads impregnated with biocrust biota. Initial viability tests showed that multiple organismal groups retained viability through the bead-preparation process, but this method has not yet been trialed in the field.

Remaining challenges & opportunities

Early biocrust pellet trials have spanned multiple ecosystems and experimental contexts, making it challenging to identify what differentiates successful deployments from failures. Differences in the traits of biocrust biota merit investigation as predictors of pelletization benefits. The moss *S. ruralis* benefitted from inclusion in some pellets, perhaps because its large size and strong morphological changes upon wetting may break open pellets upon hydration (Rosentreter et al., 2006; Zander and Eckel, 2007). Taxa readily establishing from leaf fragments, including *Syntrichia* spp. (Liu et al., 2021), may be primed for pellet benefits, though regeneration can be sensitive to fragment size (Hurtado et al., 2022). Since

only *B. argenteum* among three fire moss species established from diatomaceous earth pellets (Grover et al., 2022), and this species propagates from specialized asexual propagules (Horsley et al., 2011), gemmae or tuber production may also predict pelletization compatibility. Lichens and cyanobacteria have very different traits than these mosses, and will likely have unique determinants of pellet compatibility. Systematic, multi-site, multi-species trials of identical pellets could test these trait hypotheses.

If different settings and/or incorporation of different taxa yield different outcomes, whether one best pellet type exists becomes questionable. Different pellet types may prove optimal in different settings. Systematic tests could be used to create production and use guidelines for various applications, ideally identifying general principles for optimizing biocrust pellets per setting. Even within single projects, using multiple pellet types (e.g., fast- and slow-release combinations) may prove most successful through bet-hedging, increasing likelihood that some materials will respond to different advantageous establishment conditions (Rader et al., 2021).

Biocrust-vascular plant interactions are complicated and context-dependent (Havrilla et al., 2019), creating potential for diverse positive and negative outcomes when restoring both simultaneously. Co-delivered biocrust and seed may support mutual establishment or alternatively inhibit each other. Mixed interactions will likely occur, with interaction nature shifting based on community members and environmental conditions. Dadzie et al. (2022) observed this when seed pellets inoculated with cyanobacteria and/or heterotrophic bacteria were planted at a mine site. Some bacteria promoted seedling emergence while others showed

no effect, and cyanobacteria decreased survival in one plant species. Improved prediction can help determine whether biocrust-seed pellet combinations will be synergistic or antagonistic.

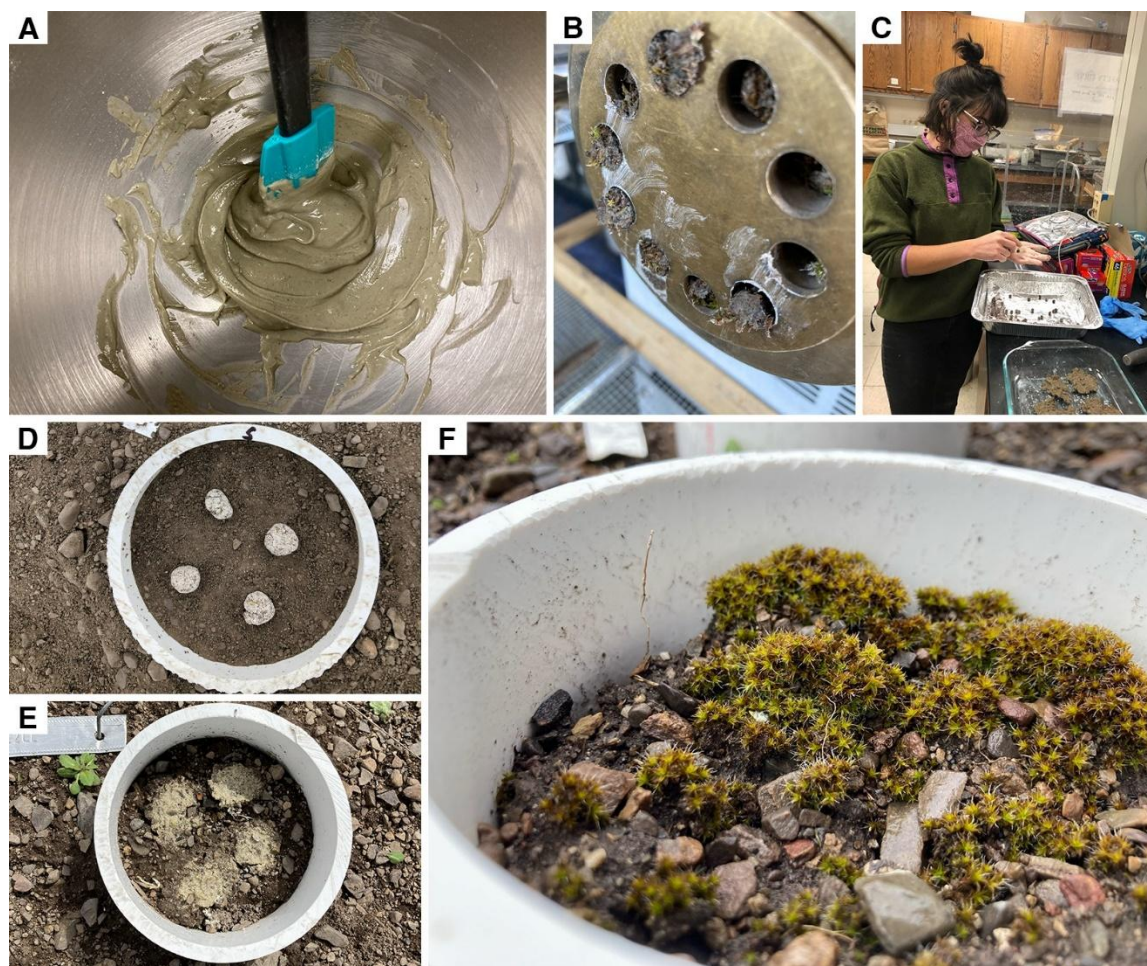


Figure 4. Manufacture and application of biocrust pellets. A. A clay-based pellet “batter” being mixed before it is formed into pellets; B. Biocrust pellets being forced through a pasta extruder. The extruded pellets can then be cut to the desired length by hand; C. Hand-forming biocrust pellets. Hand-forming allowed for greater control over the pellet shape and for the testing of multiple pellet types in early experiments. This step could plausibly be automated for greater efficiency; D. Pellets in situ immediately after deployment in western Montana; E. Pellets disintegrating after six months from fall through spring in western Montana; F. Successful

cultivation of *S. ruralis* from pellets 2.5 years after deployment in western Montana. Photo credit: Jasmine Anenberg.

Biocrust capsules

History

Biocrust capsules are the newest technique reviewed here, developed to improve upon some of the shortcomings of biocrust pellets. One shortcoming is that biocrust tolerates long term storage only when dry; storing wet or hydrating in the dark leads to loss of carbon and biomass, negatively affecting viability (Chen et al., 2024; Coe et al., 2012; Guo et al., 2020; Johnson et al., 2012). Because the current clay-pelletization process requires biocrust inoculum to be mixed with wet clay, hydrating it and embedding it into a dark environment, pelletization could degrade inoculum (we note that light transmitting pellets could plausibly avoid this problem; Oliveria et al., 2026). The need for binding agents in pellets also limits the relative volume of propagules that can be included. To address these issues, researchers began exploring the use of medical pill capsules as an alternative method of co-packaging biocrust inoculum, seeds, and additives. To our knowledge, there is no peer-reviewed literature about this method, although a similar use of “seed pills” employed by guerilla gardeners is documented in online sources such as blogs (e.g. The Subversive Gardener, Seed Pills for Guerrilla Gardening). Capsules can be inexpensive, biodegradable, and can be filled with dry ingredients. Because binders are not needed, the amount of inoculum included can be maximized. Multiple manufacturers produce a variety of capsule sizes and materials, allowing for fine tuning of capsule properties. Capsule filling systems can range from low-cost hand-

operated devices that produce small batches (e.g., 100) at a time, to automated systems used by the pharmaceutical industry for mass production. Hand production with manual machines can be challenging, depending on specific capsule contents, but with further refinement, automated production might allow for the rapid production of consistent biocrust capsules. The relative ease of creation, customizability and low cost could allow for widespread and versatile uses. Like pellets, biocrust capsules may deliver biocrust inoculum (**inoculation**), increase inoculum residence time or soil adherence (**retention**), and include additives that enhance resource access or blunt stressors (**amelioration**).

Recent developments & innovations

In the past 2 years, members of this author group have implemented a variety of studies with the aims of identifying how biocrust capsule formulation influences biocrust establishment success, and how establishment varies by: 1. differences between capsule materials, 2. differences between inoculum biocrust source, including the community makeup and the soil texture, 3. effects of different additives on establishment success and inoculum properties, and 4. the inclusion of vascular plant seeds. This research effort has been spearheaded by students (graduate, undergraduate, and high school).

A series of greenhouse-based biocrust capsule experiments explored these factors, testing seven capsule material types, five biocrust sources including greenhouse cultivated biocrusts and field collected biocrusts, and the addition of native seeds (Mayorga et al., 2026). In these experiments, biocrust capsules broke down with exposure to water, releasing biocrust biota and inducing their establishment. Capsule material strongly affected breakdown rate. Vegetable cellulose (fast) and tapioca capsules (slow) were both promising and complementary

in their breakdown rate, with vegetable cellulose capsules breaking down 2.75–3 times faster than tapioca, allowing the possibility of using both together to increase the probability of breakdown of at least some of the capsules coinciding with favorable conditions, a form of bet-hedging (Rader et al., 2021). The biocrust inoculum type (varying in source, composition, and soil texture) also influenced the rate at which the capsule materials dissolved or broke open. For example, moss-dominated inoculum appears to break through capsule walls more quickly, plausibly due to strong expansion upon wetting.

After promising greenhouse results, additional trials addressed the value of anti-herbivory and other adjunct ingredients. Anti-herbivory additives (diatomaceous earth, bergamot oil, coyote urine, pectin, habanero powder) did not change the rate of capsule breakdown in one site, but this test was also inconclusive regarding herbivory deterrence because loss of capsules to herbivores was minor (Mayorga et al., 2026). In the greenhouse, germination of vascular plants was not inhibited by encapsulation with biocrust, however timing of germination was delayed compared to dispersed unencapsulated seed (Mayorga et al., 2026). For example, Olney (2025) tested biocrust-seed capsules, also containing diatomaceous earth, in a degraded field site for two growing seasons. Encapsulated application led to greater biocrust development (cover of mosses, lichens, or dark-pigmented cyanobacteria) compared to no additions of biocrust; unencapsulated broadcast of inoculum was not clearly distinguishable from no additions. Seeding boosted the number of native plant germinants, but encapsulation neither increased nor decreased the germination rates.

Pill capsules could be used as an agent for boosting biocrust field establishment. They can provide similar protective value yet are easier and faster to produce than clay pellets, but

there are some limitations. Mosses, lichens and cyanobacteria are potentially compatible with encapsulation, as long as the propagules are in a solid state (i.e., not suspended in liquid media), are small enough to fit within the capsules, and there is a means to produce or obtain propagules. The capsule diameter limits the size, shape and traits of seeds which can be included in the capsules (e.g., awned seeds may be less compatible). Biocrust used to fill capsules must be sieved to remove rocks or organic debris that would damage the capsules while filling. The biocrust capsules must also be carefully stored to remain dry so they do not begin dissolving and clump together before deployment. While this is also the case for clay pellets, the issue is more pronounced for biocrust capsules.

Remaining challenges & opportunities

Biocrust establishment and thus biocrust capsule efficacy is dependent on numerous environmental factors, such as rainfall, which are highly unpredictable particularly under climate change (Pascale et al., 2019; Seager et al., 2023; Tan et al., 2023). Multi-year monitoring in multi-site studies is likely needed to begin to understand the utility of biocrust capsules in restoration projects. Experiments simulating different climate and weather scenarios could add yet another dimension, illuminating what *could* have happened under different conditions (e.g., simulated wet and dry years).

Possibly the greatest challenge regarding biocrust capsules is the vast number of possible variations, and one type of capsule may not be universally useful. Every site and project has unique challenges that could inhibit biocrust establishment. Rather than a universal biocrust capsule for all settings, a universal process could be used to select among the different capsule permutations and matching one or more of them to a project. We can envision a set of

guidelines in the form of a decision tree, based on a systematic series of field trials confronting capsule variations with site-specific establishment barriers. These trials could also include explicit tests of bet-hedging, comparing applications of multiple types of capsules together versus the individual types alone.



Figure 5. Manufacture and testing of biocrust capsules. A. Producing capsules in batches of 100 using a capsule filling machine. Empty capsules can be filled with sieved biocrust inoculum, native seeds, and any other desired additives. Automated capsule filling equipment exists and could greatly speed production; B. Capsules filled with biocrust inoculum and a variety of herbivory deterrents being prepared for field deployment; C. Biocrust capsules immediately after being deployed in the Mojave Desert; D. Biocrust capsules disintegrating after repeated

wetting in a greenhouse trial. The capsules shown here are a mix of various plant- and gelatin-based capsules which displayed different rates of capsule breakdown; E. Capsules with various water-holding additives disintegrating and dispersing their contents after repeated wetting in a greenhouse trial. Photo credits: Keven Griffen (A), Madeline Mayorga (B, D, & E), Lydia Bailey (C).

Biocrust slurries

History

Biocrust slurries (slurry) are created by mixing fragments of biocrust, often with adherent soil or other solids, with water or another solution to create a semi-liquid inoculum. Here, we distinguish slurries from biocrust biota applied in a liquid suspension because of some of the emergent properties of introducing more solid materials and additives into a suspension of greater viscosity. The concept of inoculating with biocrust slurries was first introduced by St. Clair et al. (1986), who used biocrust inoculum salvaged from unburned areas mixed into distilled water (250g biocrust L⁻¹), with the goal of quickly stabilizing landscapes after wildfire in central Utah. Their slurry promoted cyanobacterial recovery, and lichen recovery to a degree, but did not promote eukaryotic algae after 1 year. Maestre et al. (2006) directly compared the efficacy of slurry inoculation to addition of discrete biocrust fragments in a growth chamber, also manipulating resource availability through inclusion of sewage sludge and varying watering frequency. They found that inoculating with slurry was more successful than adding dry fragments, and that the concurrent addition of biocrust slurry, watering, and fertilization (in the form of composted sludge) had even larger success rates in increasing chlorophyll *a*, CO₂

exchange, and nitrogen fixation. Condon and Pyke (2016) experimented with mixing mosses into slurries and found regular success from year to year and establishment was found to be greatest when combined with habitat amelioration treatments such as jute net. Bu et al. (2018) developed a method of slurry application resembling hydroseeding in which pulverized moss fragments, wood fiber, and a binder were all mixed into water and applied with a mechanical sprayer into experimental field plots, in a post-agricultural semi-arid ecosystem. Both slurry application and broadcast of moss biocrust fragments resulted in high biocrust cover, but the authors did indicate some ameliorative properties of the slurry (namely, water retention and soil stabilization) compared to the broadcast method that might have resulted in faster results. Like others, Bu et al. (2018) also found that additional habitat ameliorations (shading or nutrient addition) were strong factors in aiding biocrust establishment regardless of the application method. The above examples indicate that mosses, lichens and cyanobacteria are potentially compatible with slurries, as long as there is a means to produce or obtain propagules for these biota. These early studies served to illustrate that in addition to being a method for delivery of biocrust propagules to a site (**inoculation**), slurries also provide a means to adhere inoculum to the soil surface or at least create larger inoculum-infused aggregations that are more resistant to transport (**retention**), all while being a possible vehicle for additional substances meant to improve the habitat (**amelioration**), such as nutrients, wood fiber, or polymeric binders and gelling agents.

Recent developments & innovations

Like Bu et al. (2018), some recent field studies have tested slurries versus dry inoculum, plus various habitat amelioration treatments (Ficko et al., 2023; Slate et al., 2020). Generally,

habitat amelioration treatments overwhelm any effect of biocrust delivery method. Ficko et al. (2023) found habitat ameliorations such as erosion control blanket or jute net effectively promoted arctic lichen biocrust recolonization in mining-disturbed environments, whereas slurried or dry inoculum was far less influential. Likewise, Slate et al. (2020) reported that applying jute net to bare ground before adding clay-enriched slurry or dry inoculum significantly accelerated lichen and moss growth compared to untreated plots, but both established poorly without jute net. From these studies, we can hypothesize that although slurries can improve retention and ameliorate habitat, their efficacy may be enhanced further when combined with additional habitat amelioration treatments.

Some recent research has sought to improve slurry retention and amelioration effects through varying composition. Tackifiers have particular potential as slurry additives, and multiple studies have investigated their utility in slurries and other effects on biocrust inoculum. For instance, in arid, post-mining ecosystems, Shultz et al. (2022) found that slurry dilution ratios can be quite low (1:100 vs 1:10 performed similarly) and that adding psyllium husk powder (a plant-derived tackifier) into slurries creates a mucilage that binds the soil surface, providing a cohesive substrate for biocrust development and potentially ameliorating habitat for longer-term establishment. Among multiple common hydroseeding tackifiers, exposure of biocrust moss fragments to psyllium had neutral to positive effects on moss growth (dosage dependent), while guar gum and polyacrylamide had neutral to negative effects (Blankenship et al., 2020). Broader literature supports psyllium's value in promoting biocrust recovery, either as additional treatment or as an additive in liquid media inoculants (Fick et al.,

2019; Alameda-Martín et al., 2024, Rowe et al., 2025), making its use as a slurry tackifier promising.

Remaining challenges & opportunities

Slurries show promise for biocrust restoration, but a significant current challenge is a relative lack of research about this approach. Collectively we are only scratching the surface of possible variations in slurry composition, and we lack studies that incorporate multiple slurry variations or test slurries across multiple sites. Beyond incorporation of tackifiers like psyllium (Shultz et al., 2022), nutrient sources like composted sewage sludge (Maestre et al., 2006), or water-retentive substances such as wood fiber (Bu et al., 2018), what additional enhancements can be made, and how can they be combined for maximal effect?

Two factors seem to limit the scale at which biocrust slurries might be valuable: the source and supply of biocrust inoculum and the means of dispersal and application. Most studied biocrust slurries have been created using field-collected natural biocrusts. However, inoculum produced using various biocrust cultivation technologies could be used to create slurries, expanding the treatable area. Some benefits of slurry techniques may be observed using cultivated rather than field-collected biocrusts.

Slurries would ideally be produced locally with dry ingredients mixed with water near the application site, shortly before use. Doing so could avoid some of the costs of transporting heavy loads over long distances, and could decrease the amount of time that biocrust biota remained active in the absence of light (e.g., within opaque containers). Slurry application with ground-based machinery seems most plausible, whereas aircraft dispersal could be technically feasible (e.g., slurry bombers), but potentially cost-prohibitive. Ground-based equipment

already exists for semi-liquid distribution—namely hydroseeders and slurry spreaders—which could deploy biocrust slurry for medium-scaled problems. Road access for machinery may limit where slurries could be deployed. These technologies have not yet been tested in a project-scale, machine assisted application of biocrust slurries for restoration.



Figure 6. Examples of two methods for applying biocrust slurries to soil surfaces, adaptable to different scales and budget scenarios. A. Using a hydroseeder to apply biocrust slurry. B. Using a low-cost, low-technology method to disperse slurry: manual expulsion from a plastic bag. C. Hydroseeding resulted in a uniform distribution of slurry across the plot area, but reliance on machinery limits the spatial extent of applicability D. Manual expulsion from a bag created a less uniform inoculation but is easier to deploy in less accessible areas. Photo credits: Anita Antoninka (A & C), Cara Lauria (B & D).

Conclusions - Increasing and broadening the benefits of biocrust restoration

Globally, biocrusts are major Earth system players. Recent estimates suggest >40% of terrestrial N-fixation is conducted by biocrusts (Rodriguez-Caballero et al., 2018), and global dust emission would be 60% greater without biocrusts (Rodriguez-Caballero et al., 2022). Locally, biocrust loss leads to accelerated soil erosion, fertility loss, and exotic plant invasions. A need for biocrust restoration methods has been recognized since the 1980s (St. Clair et al., 1986), with much innovation since, especially in production of biocrust inoculum. However, it has proven challenging to achieve successful assisted recovery of biocrusts using human-produced inoculum, and some of the most successful examples with regards to speed or spatial scale have required inputs like irrigation that would not be feasible in most situations, either due to resource, financial, or infrastructural limitations (Liu et al., 2013; Zhao et al., 2016; Zhou et al., 2020). An oft-overlooked element of biocrust restoration is application methods for field inoculum. Selection of proper application techniques might be the most influential factor determining success, and a way to reduce resource intensity and broaden use cases. This review highlights approaches integrating **inoculation-retention-amelioration** in single applications, requiring no additional inputs.

Biocrust restoration has been described as limited in capacity, largely because of limitations in the quantity of biocrust propagules that can be produced (i.e., "a scaling barrier" Heredia-Velásquez et al., 2023); this is particularly true for lichen production. One partial solution is expansion of the spatial scale of biocrust propagule production. The global push towards sustainable energy production may align well with the need for expanded biocrust propagule production. Supplying the world's electricity demand through solar power may

require roughly 0.3% of the Earth's surface, or about 38 million hectares (Victoria et al., 2021). The shade produced by solar panels can facilitate biocrust propagation, as shade may enhance biocrust establishment (Antoninka et al., 2018). Thus, the opportunity exists to scale up some types of biocrust propagation using solar power station shade (Heredia-Velázquez et al., 2023), perhaps in the form of the products and application methods we review here. Technical and efficiency barriers limit some biocrust propagule production methods. Though production ceilings of biocrust materials are real current limitations, we can do more with the propagule supply that we can produce. One strategy to do more with limited inoculum production capability is maximizing post-inoculation establishment probability. If survivorship and establishment can be doubled (as in Doherty et al., 2020b, Román et al., 2021, Bowker et al., 2025), treatable area increases proportionally. The methods in this review advance this goal by boosting survivorship and establishment. A second approach is strategic spatial placement to maximize outcomes. Restoration islands concentrate resources on high-quality habitat patches within degraded areas rather than spreading effort evenly (Hulvey et al., 2017). Successful islands become propagule sources, allowing recovery to expand outward. Given biocrusts' naturally heterogeneous distribution, they could similarly serve as nuclei for broader recovery (Young et al., 2021). Strategic placement also includes creating elongated treated patches in fire-prone ecosystems—high in biocrust abundance but resistant to flammable exotic plants. Such fuel breaks, whether newly created or enhanced from existing roads, trails, or ridgelines, would constrain fire spread landscape-wide while treating only small portions (Condon et al., 2023). A final strategy is leveraging mechanized dispersal over hand labor. Tractor-draft implements like rangeland drills and imprinters could scale moss biocrust establishment

(Doherty et al., 2020a). Pelletized, encapsulated, or cloth-fixed organisms could be distributed via seed drills or drones; slurries via manure spreaders. Integrating next-generation biocrust products with mechanized techniques enables efficient distribution over larger areas.

The next generation techniques for biocrust restoration that we describe here can be used for small- to medium-scaled challenges now, and with increased production efficiency and informed, strategic use of the products, could be used for large-scaled challenges in the future. Solutions at all scales can be beneficial. Main considerations for choosing among these methods for deployment or further research include: the problem's nature and spatial extent, desired biocrust types, and road or foot access feasibility. Cloths, sods and slurries best suit targeted applications like erosional features, restoration islands or firebreaks, whereas pellets and capsules could be broadly dispersed. Cloths and sods intertwine product and inoculum production; cloths appear better for moss biocrusts, while sods can generate cyanobacteria-moss-lichen mixtures if appropriately “seeded”. For slurries, capsules and pellets, inoculum production is separate, making them theoretically compatible with various biocrust biota. However, field tests suggest greater pellet compatibility with large mosses, with limited data on slurry or capsule compatibility with mixed organismal groups. Finally, sods and cloths require hand labor and safe foot access. Ground-based vehicle access appears prerequisite for upscaling slurries, whereas lightweight pellets and capsules potentially adapt to many dispersal techniques, perhaps even from aircraft. Beneficial outcomes of biocrust restoration can be enhanced by the methodologies used to establish biocrust propagules. These techniques describe a few applications of **inoculation-retention-amelioration** to help solve the challenge

of enhancing biocrust recovery under high abiotic stress, without relying on ongoing resource inputs.

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Author Contribution Statement

MB developed the idea for this review with input from SR, JA, LB, AA, CT. CL, CT, JA, KD, Griffen, LB, LC, MB, MM, PR, and SG drafted sections. Griffen, Gibson, MB and LB conceptualized and prepared figures. LB and MB were the lead editors. All authors edited the manuscript and/or contributed original ideas.

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The authors declare no conflicts of interest.

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This review paper does not present original data.

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Graphical Abstract

